TECHNICAL PRESENTATION | CO1 – AT3

Operating System in a Banking System

- Title: OS for thousands of simultaneous transactions
- Name: Hari Charan P (192511107)
- Course: CSA0405 Operating System
- Guided By:Dr K Jayasakthi Velmurugan

Banking Workload Requirements

- **Concurrency** – thousands of deposits, withdrawals and transfers at once
- **Availability** – 24×7 service with fault tolerance
- **Security** – authentication, access control and audit logs
- **Performance** – fast response time and high throughput

Types of OS Suitable

- **Multiprogramming / time-sharing** – many users share the CPU
- **Multiprocessor OS** – parallel processing, higher throughput
- **Distributed OS** – branches, ATMs and data centres act as one system
- **Real-time (soft) OS** – bounded response for ATM and payments
- **Batch OS** – end-of-day interest, reports and reconciliation

Role of System Calls

Category	Examples	Use in banking
Process control	fork(), exec(), wait(), exit()	Create a handler per transaction
File management	open(), read(), write(), close()	Account records and logs
Device management	ioctl(), read(), write()	Disk and network card access
Information	getpid(), time()	Timestamps for audit trail
Communication	socket(), send(), recv()	ATM and client requests
Protection	chmod(), setuid()	Access control on data

Transaction Processing Using System Calls

1. **Request arrives** – recv() on a socket
2. **Handler created** – fork() / clone()
3. **Record locked** – flock() or semaphore
4. **Balance read** – open(), read()
5. **Update committed** – write(), fsync() for durability
6. **Reply and cleanup** – send(), close(), exit()

System calls are the safe gateway between banking apps and the kernel.

THANK YOU


Operating System In a Banking System
GPS Map Camera
Chennai,Tamil Nadu,India
Sivan Koil Street, Kuthambakkam, Chennai, Tamil Nadu 600124, India
Lat 13.046152, Long 80.000446
Wednesday, 07/10/2026 01:54 PM GMT+05:30
Note : Captured by GPS Map Camera